Bilderwand – layout candidates

Target 180x120 cm ± 10.0 cm, nominal gap free in [1.5, 6.0] cm, orientation fixed, sizes are measured outer dimensions. Max. gap deviation = largest difference between an actual gap and the layout's nominal gap; notches = parts of the outer border not covered by a frame or a regular gap. Ranked by a combined score (gap uniformity first, border sharpness second, then number of frames).

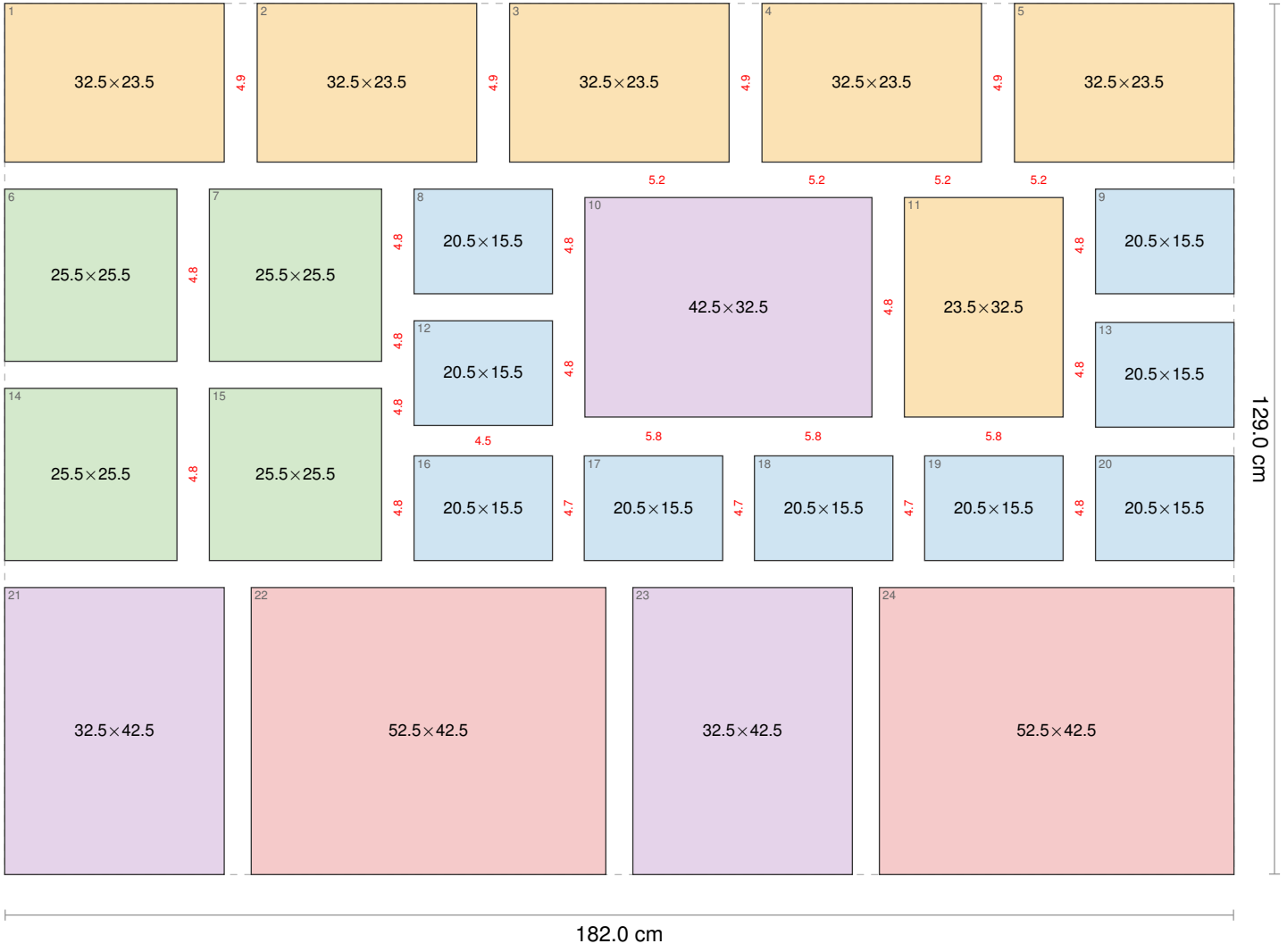
#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	182.0 × 129.0	4	24	1.8	0.56	none	4/4	0.65
2	184.0 × 115.0	2	20	2.0	0.65	4 cm	4/4	0.75
3	179.5 × 120.5	1.5	26	2.5	0.58	10 cm	4/4	0.67
4	172.5 × 125.5	2.5	25	3.5	0.70	none	4/4	0.75
5	174.5 × 130.5	1.5	25	2.5	0.95	6 cm	4/4	0.78
6	172.5 × 125.5	2.5	25	3.8	0.69	none	4/4	0.78
7	179.5 × 111.5	1.5	25	4.0	0.51	none	4/4	0.79
8	181.5 × 129.5	3.5	24	3.0	0.64	6 cm	4/4	0.77
9	176.5 × 113.5	1.5	23	3.5	0.70	none	4/4	0.79
10	182.5 × 129.5	3.5	23	4.0	0.86	none	4/4	0.82
11	184.5 × 113.5	1.5	23	3.5	0.86	4 cm	4/4	0.78
12	177.5 × 118.5	2.5	22	3.0	0.83	4 cm	4/4	0.72
13	173.5 × 116.5	3	21	3.0	0.63	6 cm	4/4	0.79
14	176.0 × 125.0	2	20	3.0	0.91	4 cm	4/4	0.77
15	179.5 × 120.5	1.5	26	3.0	0.61	8 cm	4/4	0.77
16	179.5 × 120.5	1.5	26	2.5	0.60	13 cm	4/4	0.77

Inventory

	outer w × h (cm)		have
	20.5×15.5	L	9
	15.5×20.5	P	2
	32.5×42.5	P	2
	42.5×32.5	L	1
	32.5×23.5	L	5
	23.5×32.5	P	1
	25.5×25.5		5
	42.5×52.5	P	1
	52.5×42.5	L	2

Search parameters: target = 180x120, tol = 10.0, gap_min = 1.5, gap_max = 6.0, gap_mode = free, delta = 0.5, slack_rule = even, cost_shape = huber, min_frames = 10, frame_bonus = 1.5, runs = 256, iters = 20 min, seed = 0, large_area = 2000.0, large_count = 2-3, large_apart = True, large_min_dist = 70.0, entropy = 40.0

Layout 1 182.0 × 129.0 cm gap 4 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

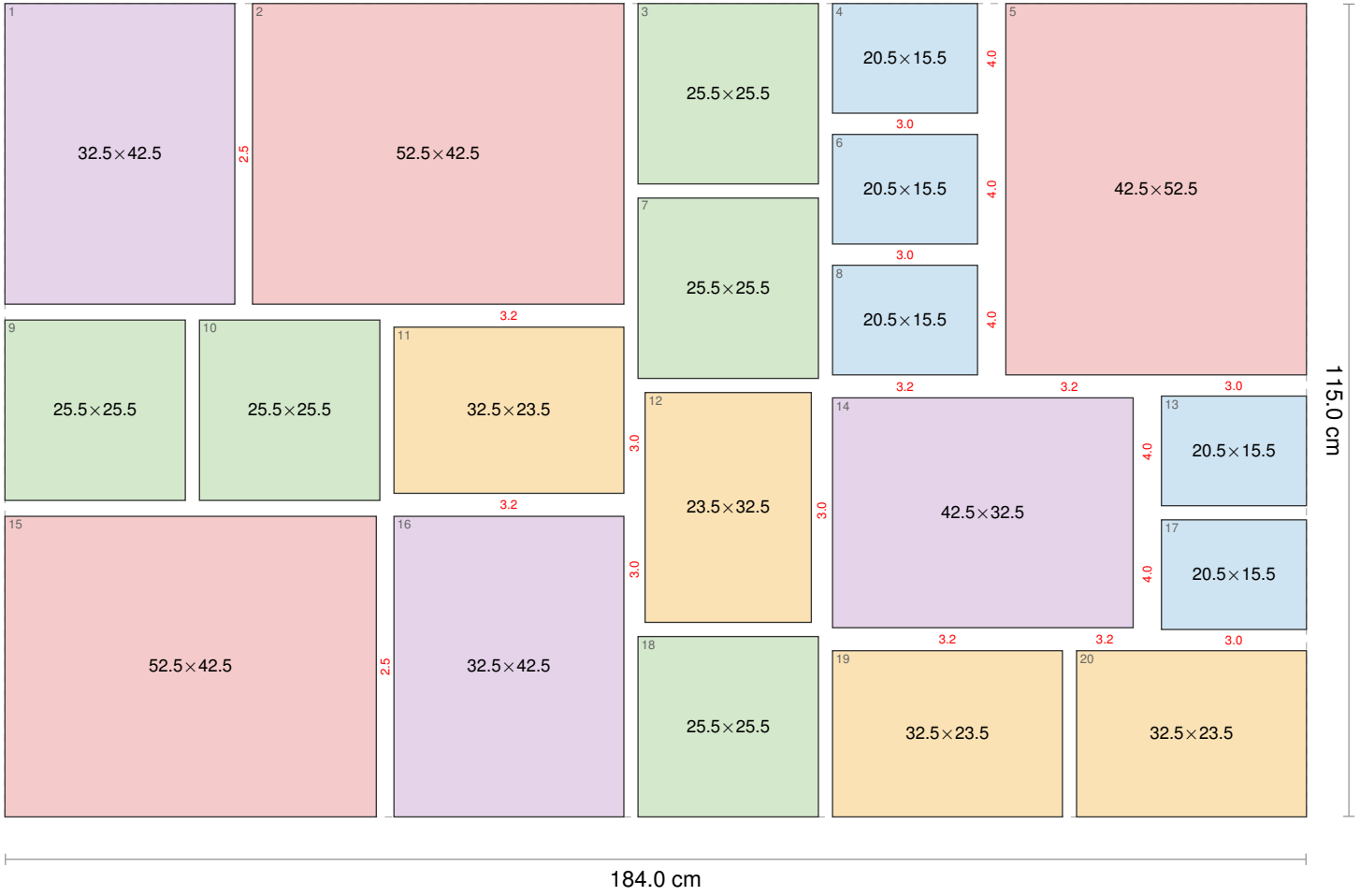
Outer size	182.0 × 129.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 5.8 cm
Max. deviation	1.8 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 93 cm
Entropy	0.65
mix / variety	0.76 / 0.60
stagger / lines	0.48 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	37.4	0.0
3	32.5 × 23.5	74.8	0.0
4	32.5 × 23.5	112.1	0.0
5	32.5 × 23.5	149.5	0.0
6	25.5 × 25.5	0.0	27.5
7	25.5 × 25.5	30.3	27.5
8	20.5 × 15.5	60.6	27.5
9	20.5 × 15.5	161.5	27.5
10	42.5 × 32.5	85.9	28.8
11	23.5 × 32.5	133.2	28.8
12	20.5 × 15.5	60.6	47.0
13	20.5 × 15.5	161.5	47.2
14	25.5 × 25.5	0.0	57.0
15	25.5 × 25.5	30.3	57.0
16	20.5 × 15.5	60.6	67.0
17	20.5 × 15.5	85.8	67.0
18	20.5 × 15.5	111.0	67.0
19	20.5 × 15.5	136.2	67.0
20	20.5 × 15.5	161.5	67.0
21	32.5 × 42.5	0.0	86.5
22	52.5 × 42.5	36.5	86.5
23	32.5 × 42.5	93.0	86.5
24	52.5 × 42.5	129.5	86.5

Layout 2 184.0 × 115.0 cm gap 2 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.0 cm
Max. deviation	2.0 cm
Mean deviation	0.65 cm
Border notches	top 4 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, not touching, min. dist. 81 cm
Entropy	0.75
mix / variety	0.86 / 0.80
stagger / lines	0.59 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	5	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	3	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.0	0.0
3	25.5 × 25.5	89.5	0.0
4	20.5 × 15.5	117.0	0.0
5	42.5 × 52.5	141.5	0.0
6	20.5 × 15.5	117.0	18.5
7	25.5 × 25.5	89.5	27.5
8	20.5 × 15.5	117.0	37.0
9	25.5 × 25.5	0.0	44.8
10	25.5 × 25.5	27.5	44.8
11	32.5 × 23.5	55.0	45.8
12	23.5 × 32.5	90.5	55.0
13	20.5 × 15.5	163.5	55.5
14	42.5 × 32.5	117.0	55.8
15	52.5 × 42.5	0.0	72.5
16	32.5 × 42.5	55.0	72.5
17	20.5 × 15.5	163.5	73.0
18	25.5 × 25.5	89.5	89.5
19	32.5 × 23.5	117.0	91.5
20	32.5 × 23.5	151.5	91.5

Layout 3 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.58 cm
Border notches	top 10 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 91 cm
Entropy	0.67
mix / variety	0.83 / 0.66
stagger / lines	0.45 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	55.7	0.0
3	52.5 × 42.5	91.3	0.0
4	32.5 × 42.5	147.0	0.0
5	25.5 × 25.5	0.0	44.0
6	25.5 × 25.5	62.7	44.0
7	25.5 × 25.5	126.2	44.0
8	25.5 × 25.5	154.0	44.0
9	32.5 × 23.5	27.8	45.0
10	32.5 × 23.5	91.3	45.0
11	20.5 × 15.5	0.0	71.0
12	20.5 × 15.5	22.0	71.0
13	20.5 × 15.5	44.0	71.0
14	20.5 × 15.5	66.0	71.0
15	20.5 × 15.5	89.5	71.0
16	32.5 × 23.5	147.0	71.0
17	15.5 × 20.5	113.0	72.5
18	15.5 × 20.5	130.0	72.5
19	42.5 × 32.5	0.0	88.0
20	20.5 × 15.5	44.0	88.0
21	20.5 × 15.5	66.0	88.0
22	23.5 × 32.5	88.0	88.0
23	32.5 × 23.5	113.0	97.0
24	32.5 × 23.5	147.0	97.0
25	20.5 × 15.5	44.0	105.0
26	20.5 × 15.5	66.0	105.0

Layout 4 172.5 × 125.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

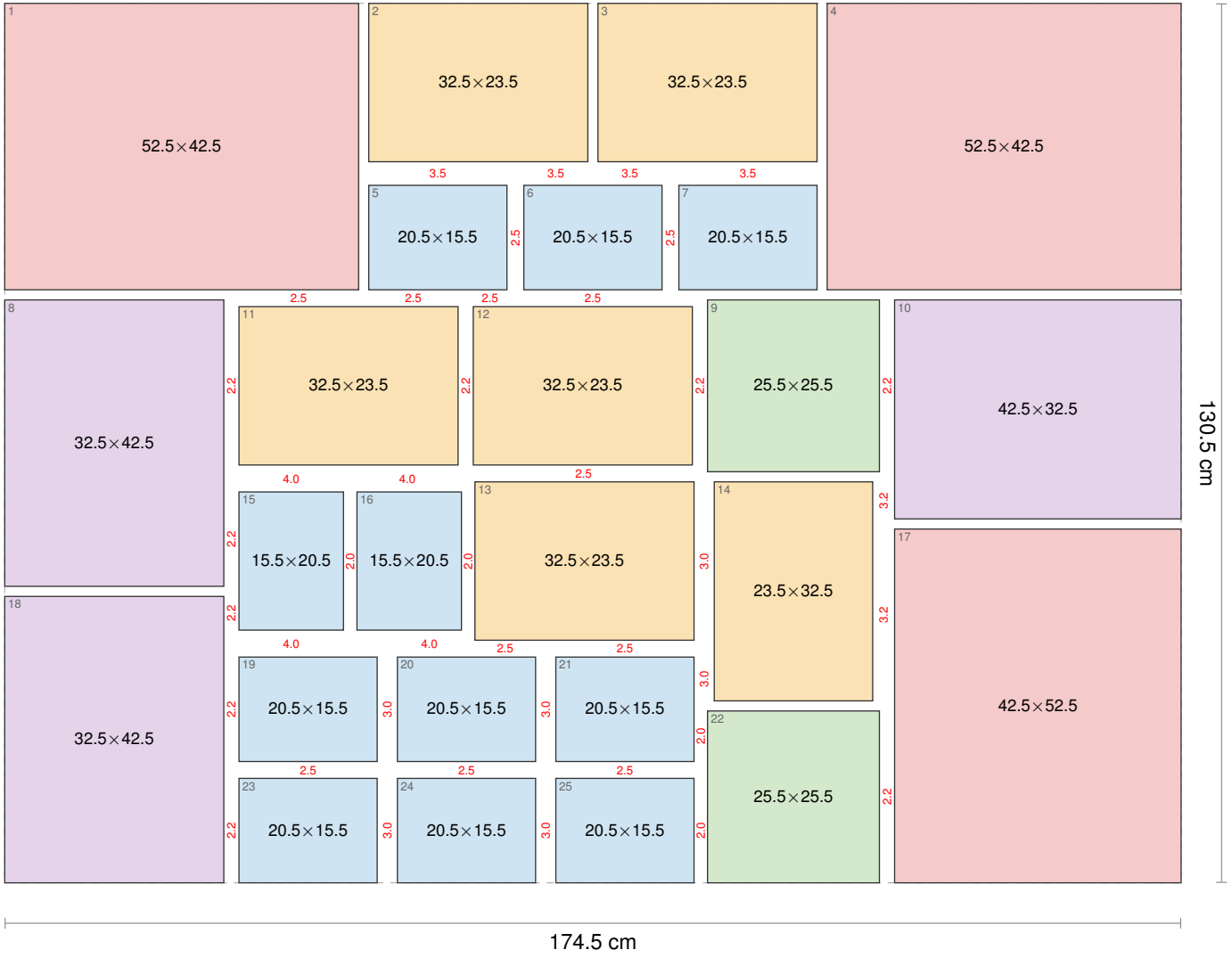
Outer size	172.5 × 125.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.0 cm
Max. deviation	3.5 cm
Mean deviation	0.70 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 80 cm
Entropy	0.75
mix / variety	0.79 / 0.73
stagger / lines	0.50 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 23.5	35.0	0.0
3	32.5 × 23.5	70.0	0.0
4	32.5 × 23.5	105.0	0.0
5	32.5 × 42.5	140.0	0.0
6	20.5 × 15.5	35.0	26.0
7	20.5 × 15.5	60.0	26.0
8	52.5 × 42.5	85.0	26.5
9	25.5 × 25.5	55.0	44.0
10	32.5 × 23.5	0.0	46.0
11	32.5 × 23.5	140.0	46.0
12	15.5 × 20.5	35.0	46.5
13	25.5 × 25.5	0.0	72.0
14	20.5 × 15.5	75.3	72.0
15	20.5 × 15.5	100.2	72.0
16	20.5 × 15.5	123.6	72.0
17	25.5 × 25.5	147.0	72.0
18	42.5 × 52.5	28.4	73.0
19	20.5 × 15.5	100.2	91.0
20	20.5 × 15.5	123.6	91.0
21	23.5 × 32.5	73.8	93.0
22	25.5 × 25.5	0.0	100.0
23	25.5 × 25.5	147.0	100.0
24	20.5 × 15.5	100.2	110.0
25	20.5 × 15.5	123.6	110.0

Layout 5 174.5 × 130.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

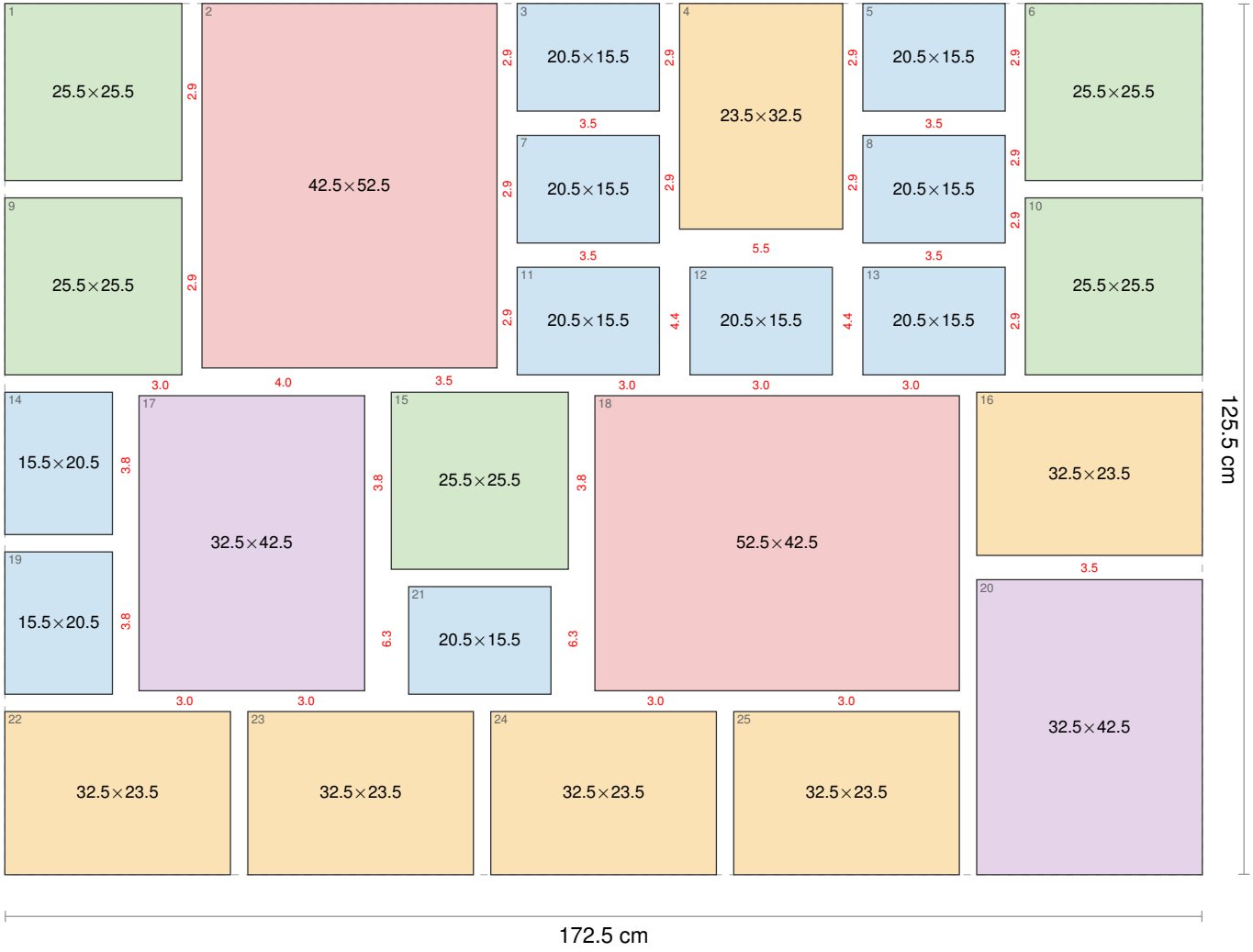
Outer size	174.5 × 130.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.95 cm
Border notches	bottom 6 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	3, not touching, min. dist. 83 cm
Entropy	0.78
mix / variety	0.86 / 0.74
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		2	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	54.0	0.0
3	32.5 × 23.5	88.0	0.0
4	52.5 × 42.5	122.0	0.0
5	20.5 × 15.5	54.0	27.0
6	20.5 × 15.5	77.0	27.0
7	20.5 × 15.5	100.0	27.0
8	32.5 × 42.5	0.0	44.0
9	25.5 × 25.5	104.2	44.0
10	42.5 × 32.5	132.0	44.0
11	32.5 × 23.5	34.8	45.0
12	32.5 × 23.5	69.5	45.0
13	32.5 × 23.5	69.8	71.0
14	23.5 × 32.5	105.2	71.0
15	15.5 × 20.5	34.8	72.5
16	15.5 × 20.5	52.2	72.5
17	42.5 × 52.5	132.0	78.0
18	32.5 × 42.5	0.0	88.0
19	20.5 × 15.5	34.8	97.0
20	20.5 × 15.5	58.2	97.0
21	20.5 × 15.5	81.8	97.0
22	25.5 × 25.5	104.2	105.0
23	20.5 × 15.5	34.8	115.0
24	20.5 × 15.5	58.2	115.0
25	20.5 × 15.5	81.8	115.0

Layout 6 172.5 × 125.5 cm gap 2.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

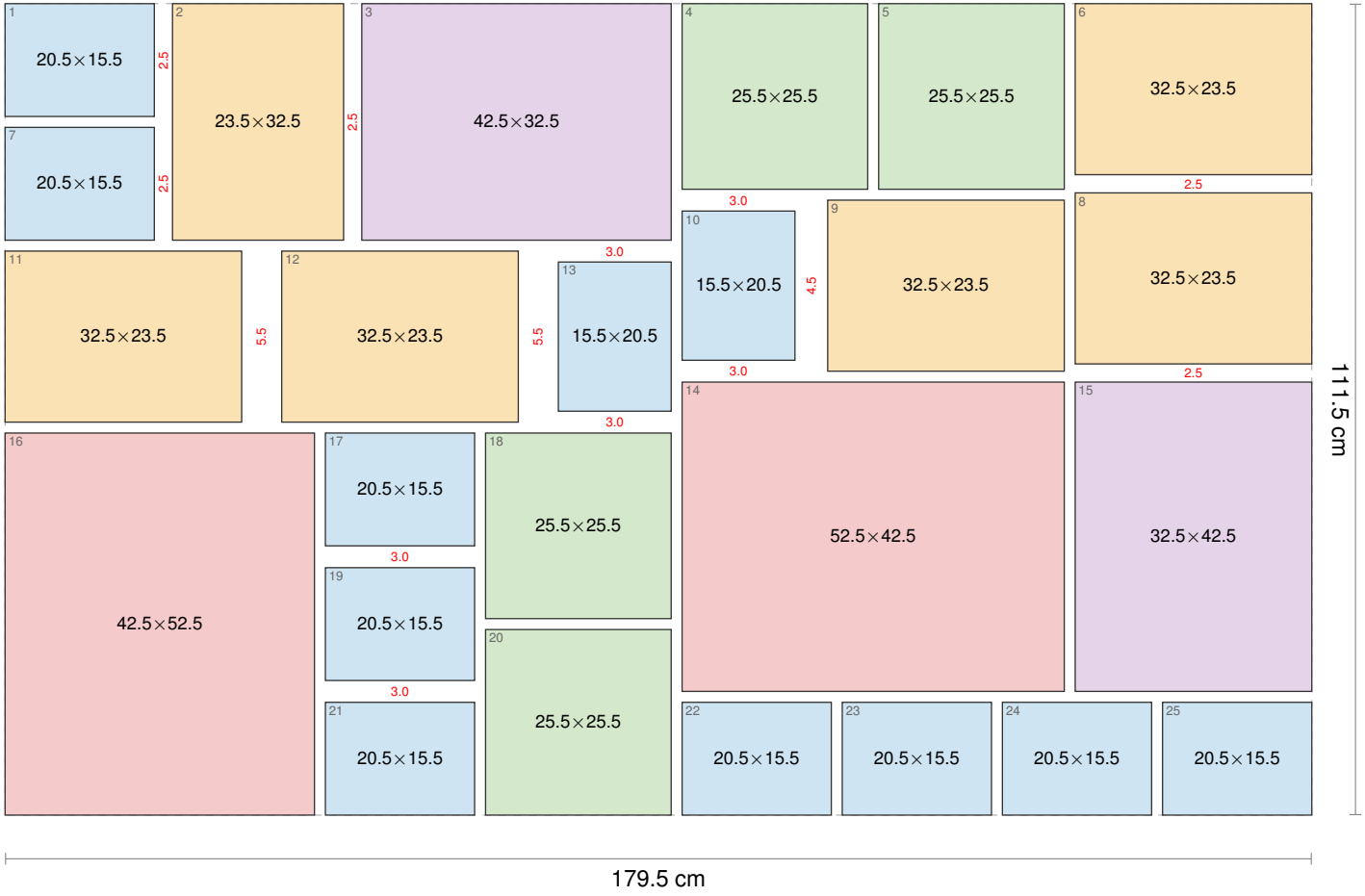
Outer size	172.5 × 125.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.3 cm
Max. deviation	3.8 cm
Mean deviation	0.69 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 80 cm
Entropy	0.78
mix / variety	0.82 / 0.77
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	28.4	0.0
3	20.5 × 15.5	73.8	0.0
4	23.5 × 32.5	97.2	0.0
5	20.5 × 15.5	123.6	0.0
6	25.5 × 25.5	147.0	0.0
7	20.5 × 15.5	73.8	19.0
8	20.5 × 15.5	123.6	19.0
9	25.5 × 25.5	0.0	28.0
10	25.5 × 25.5	147.0	28.0
11	20.5 × 15.5	73.8	38.0
12	20.5 × 15.5	98.7	38.0
13	20.5 × 15.5	123.6	38.0
14	15.5 × 20.5	0.0	56.0
15	25.5 × 25.5	55.7	56.0
16	32.5 × 23.5	140.0	56.0
17	32.5 × 42.5	19.3	56.5
18	52.5 × 42.5	85.0	56.5
19	15.5 × 20.5	0.0	79.0
20	32.5 × 42.5	140.0	83.0
21	20.5 × 15.5	58.2	84.0
22	32.5 × 23.5	0.0	102.0
23	32.5 × 23.5	35.0	102.0
24	32.5 × 23.5	70.0	102.0
25	32.5 × 23.5	105.0	102.0

Layout 7 179.5 × 111.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 111.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.5 cm
Max. deviation	4.0 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 99 cm
Entropy	0.79
mix / variety	0.83 / 0.78
stagger / lines	0.54 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	23.5 × 32.5	23.0	0.0
3	42.5 × 32.5	49.0	0.0
4	25.5 × 25.5	93.0	0.0
5	25.5 × 25.5	120.0	0.0
6	32.5 × 23.5	147.0	0.0
7	20.5 × 15.5	0.0	17.0
8	32.5 × 23.5	147.0	26.0
9	32.5 × 23.5	113.0	27.0
10	15.5 × 20.5	93.0	28.5
11	32.5 × 23.5	0.0	34.0
12	32.5 × 23.5	38.0	34.0
13	15.5 × 20.5	76.0	35.5
14	52.5 × 42.5	93.0	52.0
15	32.5 × 42.5	147.0	52.0
16	42.5 × 52.5	0.0	59.0
17	20.5 × 15.5	44.0	59.0
18	25.5 × 25.5	66.0	59.0
19	20.5 × 15.5	44.0	77.5
20	25.5 × 25.5	66.0	86.0
21	20.5 × 15.5	44.0	96.0
22	20.5 × 15.5	93.0	96.0
23	20.5 × 15.5	115.0	96.0
24	20.5 × 15.5	137.0	96.0
25	20.5 × 15.5	159.0	96.0

Layout 8 181.5 × 129.5 cm gap 3.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

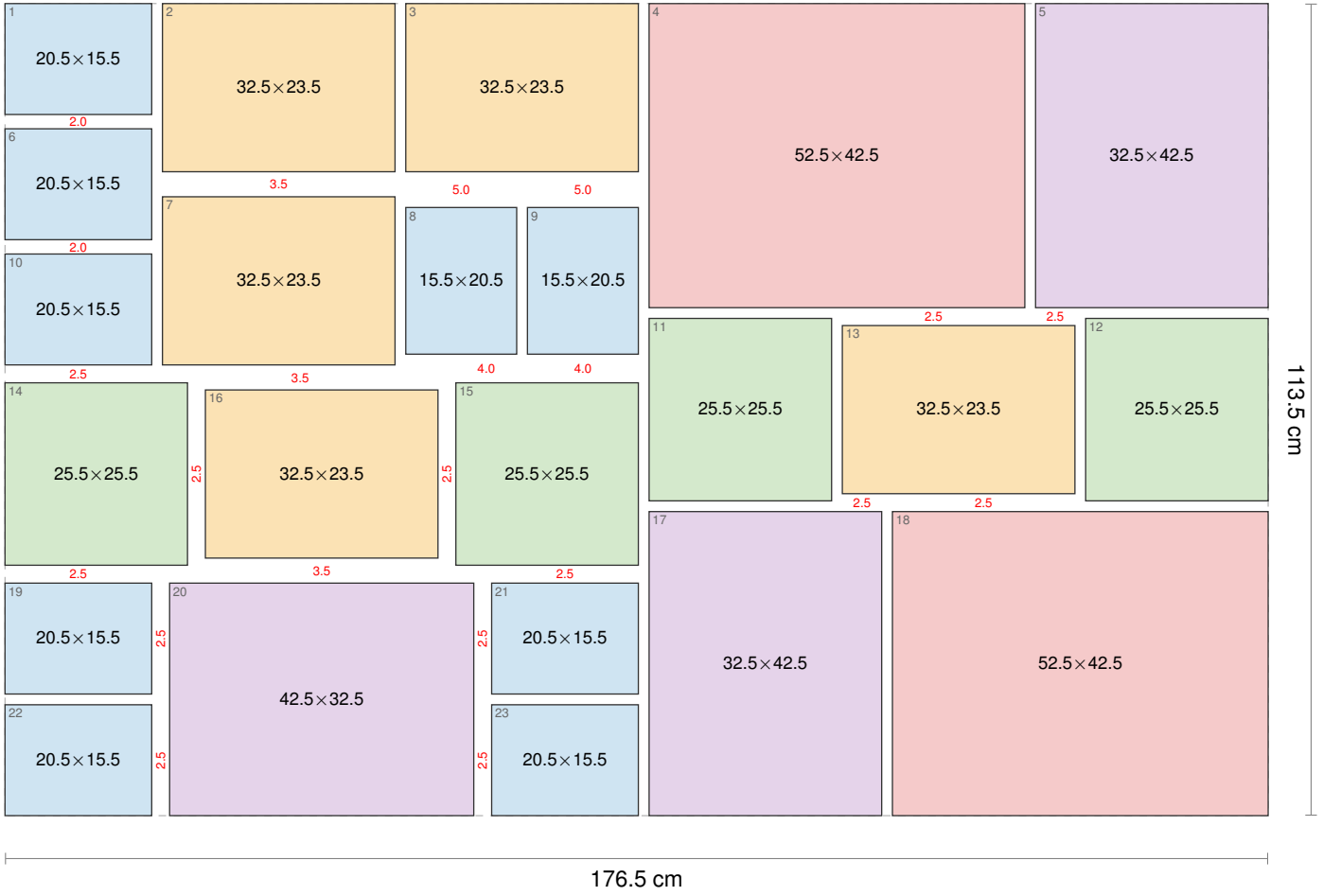
Outer size	181.5 × 129.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 6.5 cm
Max. deviation	3.0 cm
Mean deviation	0.64 cm
Border notches	right 6 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 84 cm
Entropy	0.77
mix / variety	0.76 / 0.79
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	56.5	0.0
3	52.5 × 42.5	93.0	0.0
4	32.5 × 23.5	149.0	0.0
5	32.5 × 42.5	149.0	29.0
6	25.5 × 25.5	0.0	46.0
7	25.5 × 25.5	120.0	46.0
8	32.5 × 23.5	29.0	47.0
9	32.5 × 23.5	84.0	47.0
10	15.5 × 20.5	65.0	48.5
11	20.5 × 15.5	0.0	75.0
12	25.5 × 25.5	70.8	75.0
13	20.5 × 15.5	136.6	75.0
14	20.5 × 15.5	161.0	75.0
15	32.5 × 23.5	100.2	76.0
16	42.5 × 52.5	24.4	77.0
17	20.5 × 15.5	0.0	94.5
18	20.5 × 15.5	136.6	94.5
19	20.5 × 15.5	161.0	94.5
20	25.5 × 25.5	107.2	104.0
21	32.5 × 23.5	70.8	106.0
22	20.5 × 15.5	0.0	114.0
23	20.5 × 15.5	136.6	114.0
24	20.5 × 15.5	161.0	114.0

Layout 9 176.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	176.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.70 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	2, not touching, min. dist. 79 cm
Entropy	0.79
mix / variety	0.81 / 0.82
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	32.5 × 23.5	22.0	0.0
3	32.5 × 23.5	56.0	0.0
4	52.5 × 42.5	90.0	0.0
5	32.5 × 42.5	144.0	0.0
6	20.5 × 15.5	0.0	17.5
7	32.5 × 23.5	22.0	27.0
8	15.5 × 20.5	56.0	28.5
9	15.5 × 20.5	73.0	28.5
10	20.5 × 15.5	0.0	35.0
11	25.5 × 25.5	90.0	44.0
12	25.5 × 25.5	151.0	44.0
13	32.5 × 23.5	117.0	45.0
14	25.5 × 25.5	0.0	53.0
15	25.5 × 25.5	63.0	53.0
16	32.5 × 23.5	28.0	54.0
17	32.5 × 42.5	90.0	71.0
18	52.5 × 42.5	124.0	71.0
19	20.5 × 15.5	0.0	81.0
20	42.5 × 32.5	23.0	81.0
21	20.5 × 15.5	68.0	81.0
22	20.5 × 15.5	0.0	98.0
23	20.5 × 15.5	68.0	98.0

Layout 10 182.5 × 129.5 cm gap 3.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	182.5 × 129.5 cm
Nominal gap	3.5 cm
Actual gaps	3.5 – 7.5 cm
Max. deviation	4.0 cm
Mean deviation	0.86 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 84 cm
Entropy	0.82
mix / variety	0.83 / 0.87
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	56.7	0.0
3	52.5 × 42.5	93.3	0.0
4	32.5 × 42.5	150.0	0.0
5	25.5 × 25.5	0.0	46.0
6	25.5 × 25.5	67.0	46.0
7	20.5 × 15.5	133.0	46.0
8	25.5 × 25.5	157.0	46.0
9	32.5 × 23.5	30.0	47.0
10	32.5 × 23.5	97.0	47.0
11	15.5 × 20.5	135.5	67.0
12	25.5 × 25.5	157.0	75.0
13	20.5 × 15.5	0.0	76.0
14	32.5 × 23.5	70.0	76.0
15	20.5 × 15.5	107.5	76.0
16	42.5 × 52.5	24.0	77.0
17	20.5 × 15.5	133.0	93.0
18	20.5 × 15.5	0.0	95.0
19	23.5 × 32.5	106.0	97.0
20	25.5 × 25.5	157.0	104.0
21	32.5 × 23.5	70.0	106.0
22	20.5 × 15.5	0.0	114.0
23	20.5 × 15.5	133.0	114.0

Layout 11 184.5 × 113.5 cm gap 1.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	184.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.86 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.78
mix / variety	0.77 / 0.82
stagger / lines	0.52 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	54.8	0.0
3	32.5 × 23.5	82.5	0.0
4	32.5 × 23.5	117.2	0.0
5	32.5 × 42.5	152.0	0.0
6	20.5 × 15.5	57.2	27.0
7	20.5 × 15.5	82.5	27.0
8	20.5 × 15.5	105.9	27.0
9	20.5 × 15.5	129.2	27.0
10	32.5 × 23.5	0.0	44.0
11	20.5 × 15.5	36.5	44.0
12	32.5 × 23.5	81.0	44.0
13	20.5 × 15.5	117.5	44.0
14	42.5 × 52.5	142.0	44.0
15	15.5 × 20.5	61.0	45.5
16	25.5 × 25.5	34.0	61.0
17	25.5 × 25.5	115.0	61.0
18	32.5 × 42.5	0.0	71.0
19	52.5 × 42.5	61.0	71.0
20	25.5 × 25.5	34.0	88.0
21	25.5 × 25.5	115.0	88.0
22	20.5 × 15.5	142.0	98.0
23	20.5 × 15.5	164.0	98.0

Layout 12 177.5 × 118.5 cm gap 2.5 cm 22 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

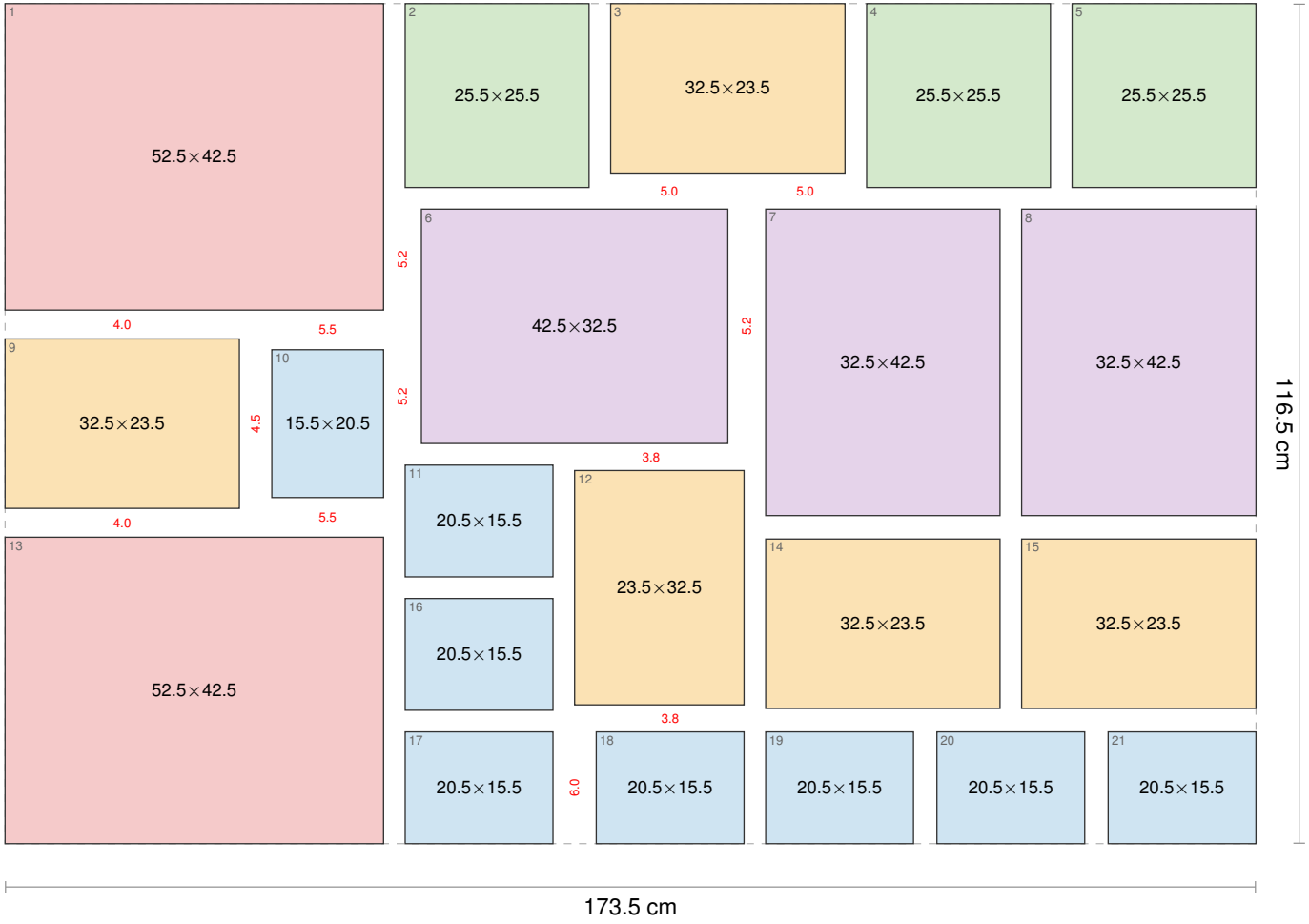
Outer size	177.5 × 118.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.83 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	22 of 28
Large frames	3, not touching, min. dist. 74 cm
Entropy	0.72
mix / variety	0.85 / 0.70
stagger / lines	0.58 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		2	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	35.0	0.0
3	32.5 × 42.5	90.0	0.0
4	52.5 × 42.5	125.0	0.0
5	15.5 × 20.5	0.0	45.0
6	15.5 × 20.5	19.0	45.0
7	20.5 × 15.5	61.0	45.0
8	20.5 × 15.5	85.0	45.0
9	20.5 × 15.5	109.0	45.0
10	20.5 × 15.5	133.0	45.0
11	20.5 × 15.5	157.0	45.0
12	20.5 × 15.5	38.0	47.5
13	42.5 × 32.5	106.5	65.0
14	25.5 × 25.5	152.0	65.0
15	42.5 × 52.5	61.0	66.0
16	32.5 × 23.5	0.0	68.0
17	20.5 × 15.5	36.5	68.0
18	23.5 × 32.5	35.0	86.0
19	25.5 × 25.5	152.0	93.0
20	32.5 × 23.5	0.0	95.0
21	20.5 × 15.5	106.0	103.0
22	20.5 × 15.5	129.0	103.0

Layout 13 173.5 × 116.5 cm gap 3 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

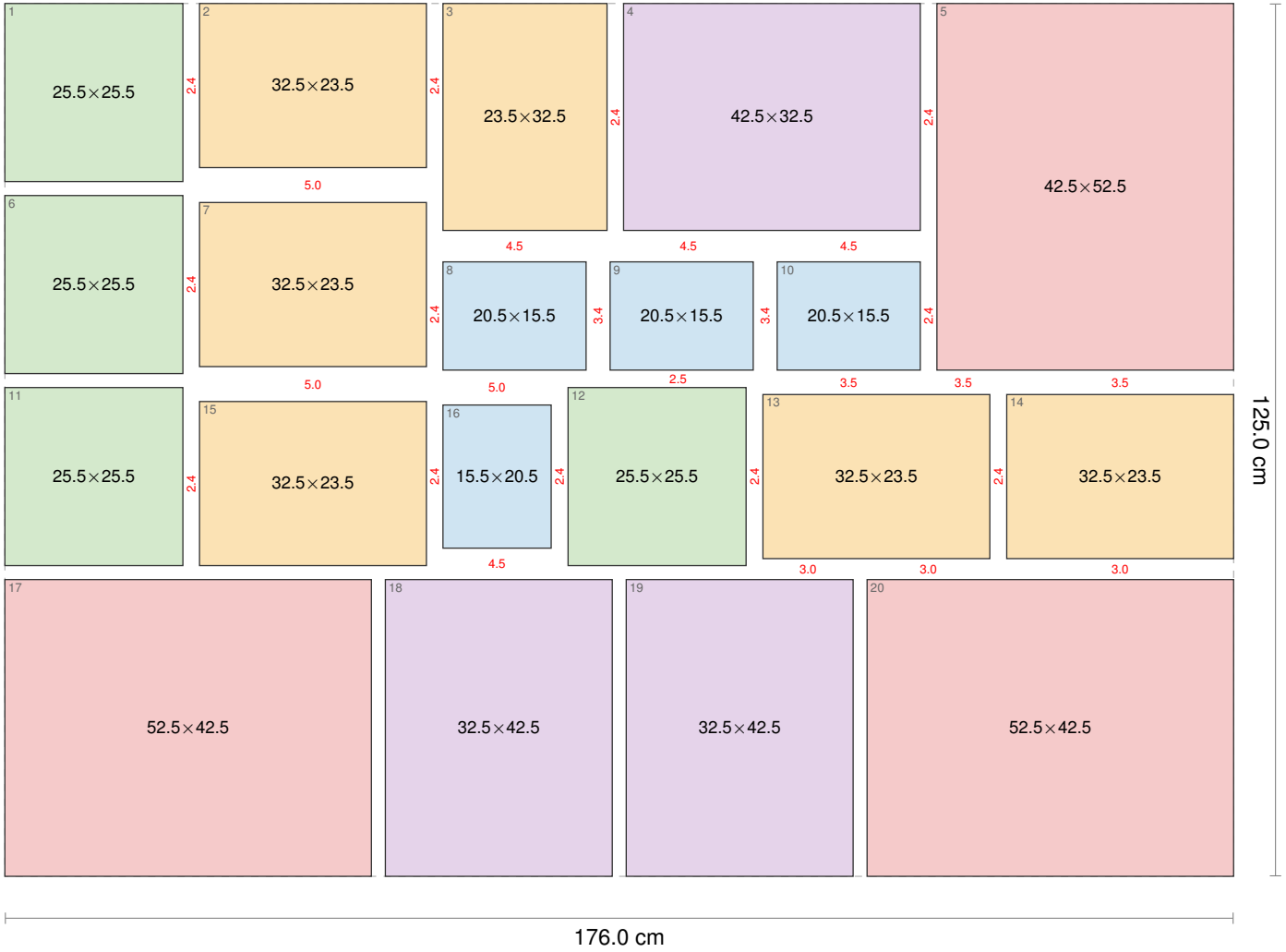
Outer size	173.5 × 116.5 cm
Nominal gap	3 cm
Actual gaps	3.0 – 6.0 cm
Max. deviation	3.0 cm
Mean deviation	0.63 cm
Border notches	bottom 6 cm
Corners filled	4 / 4
Frames used	21 of 28
Large frames	2, not touching, min. dist. 74 cm
Entropy	0.79
mix / variety	0.84 / 0.75
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	55.5	0.0
3	32.5 × 23.5	84.0	0.0
4	25.5 × 25.5	119.5	0.0
5	25.5 × 25.5	148.0	0.0
6	42.5 × 32.5	57.8	28.5
7	32.5 × 42.5	105.5	28.5
8	32.5 × 42.5	141.0	28.5
9	32.5 × 23.5	0.0	46.5
10	15.5 × 20.5	37.0	48.0
11	20.5 × 15.5	55.5	64.0
12	23.5 × 32.5	79.0	64.8
13	52.5 × 42.5	0.0	74.0
14	32.5 × 23.5	105.5	74.2
15	32.5 × 23.5	141.0	74.2
16	20.5 × 15.5	55.5	82.5
17	20.5 × 15.5	55.5	101.0
18	20.5 × 15.5	82.0	101.0
19	20.5 × 15.5	105.5	101.0
20	20.5 × 15.5	129.2	101.0
21	20.5 × 15.5	153.0	101.0

Layout 14 176.0 × 125.0 cm gap 2 cm 20 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

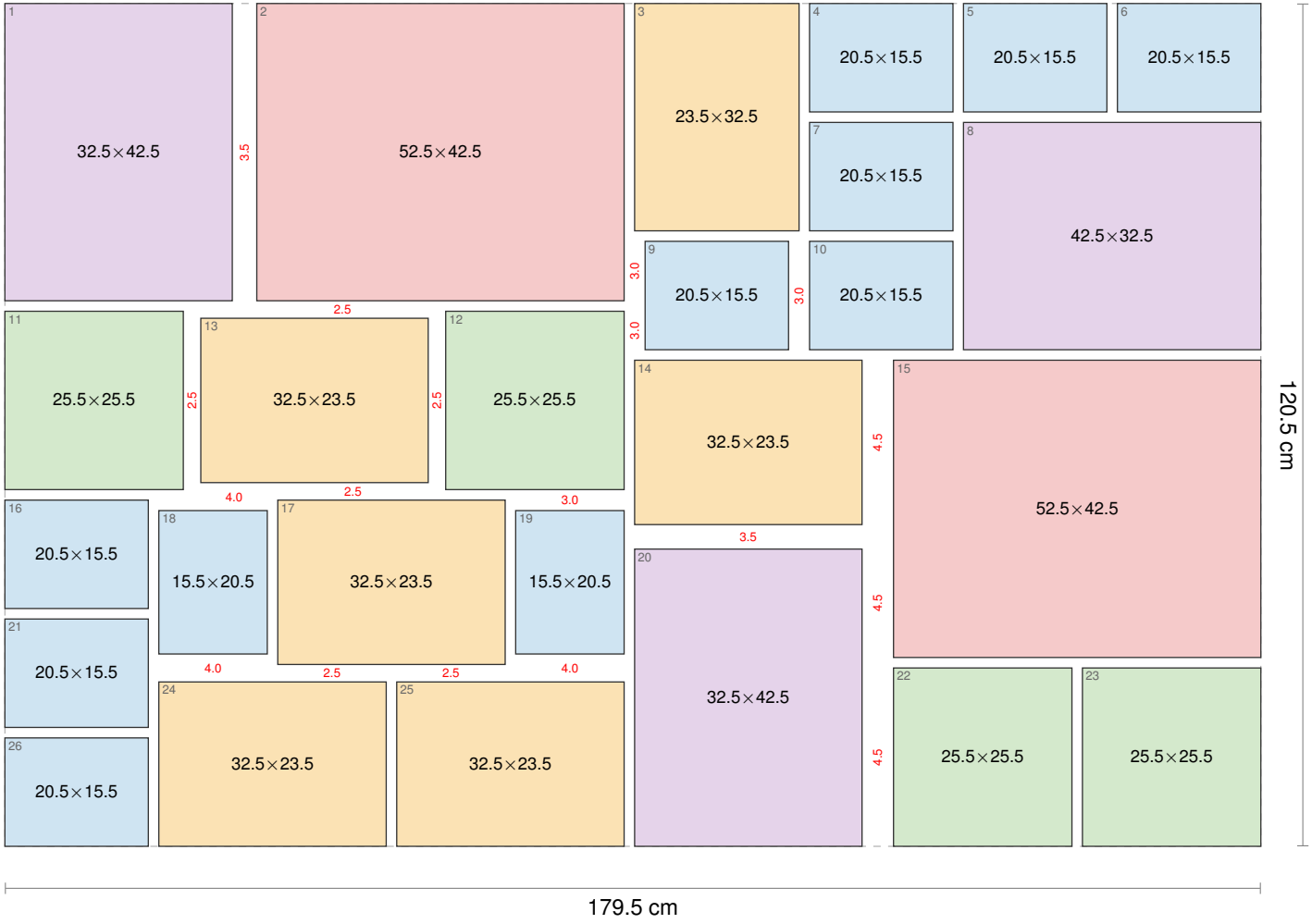
Outer size	176.0 × 125.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 5.0 cm
Max. deviation	3.0 cm
Mean deviation	0.91 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	20 of 28
Large frames	3, not touching, min. dist. 78 cm
Entropy	0.77
mix / variety	0.92 / 0.85
stagger / lines	0.56 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	3	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.9	0.0
3	23.5 × 32.5	62.8	0.0
4	42.5 × 32.5	88.6	0.0
5	42.5 × 52.5	133.5	0.0
6	25.5 × 25.5	0.0	27.5
7	32.5 × 23.5	27.9	28.5
8	20.5 × 15.5	62.8	37.0
9	20.5 × 15.5	86.7	37.0
10	20.5 × 15.5	110.6	37.0
11	25.5 × 25.5	0.0	55.0
12	25.5 × 25.5	80.7	55.0
13	32.5 × 23.5	108.6	56.0
14	32.5 × 23.5	143.5	56.0
15	32.5 × 23.5	27.9	57.0
16	15.5 × 20.5	62.8	57.5
17	52.5 × 42.5	0.0	82.5
18	32.5 × 42.5	54.5	82.5
19	32.5 × 42.5	89.0	82.5
20	52.5 × 42.5	123.5	82.5

Layout 15 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

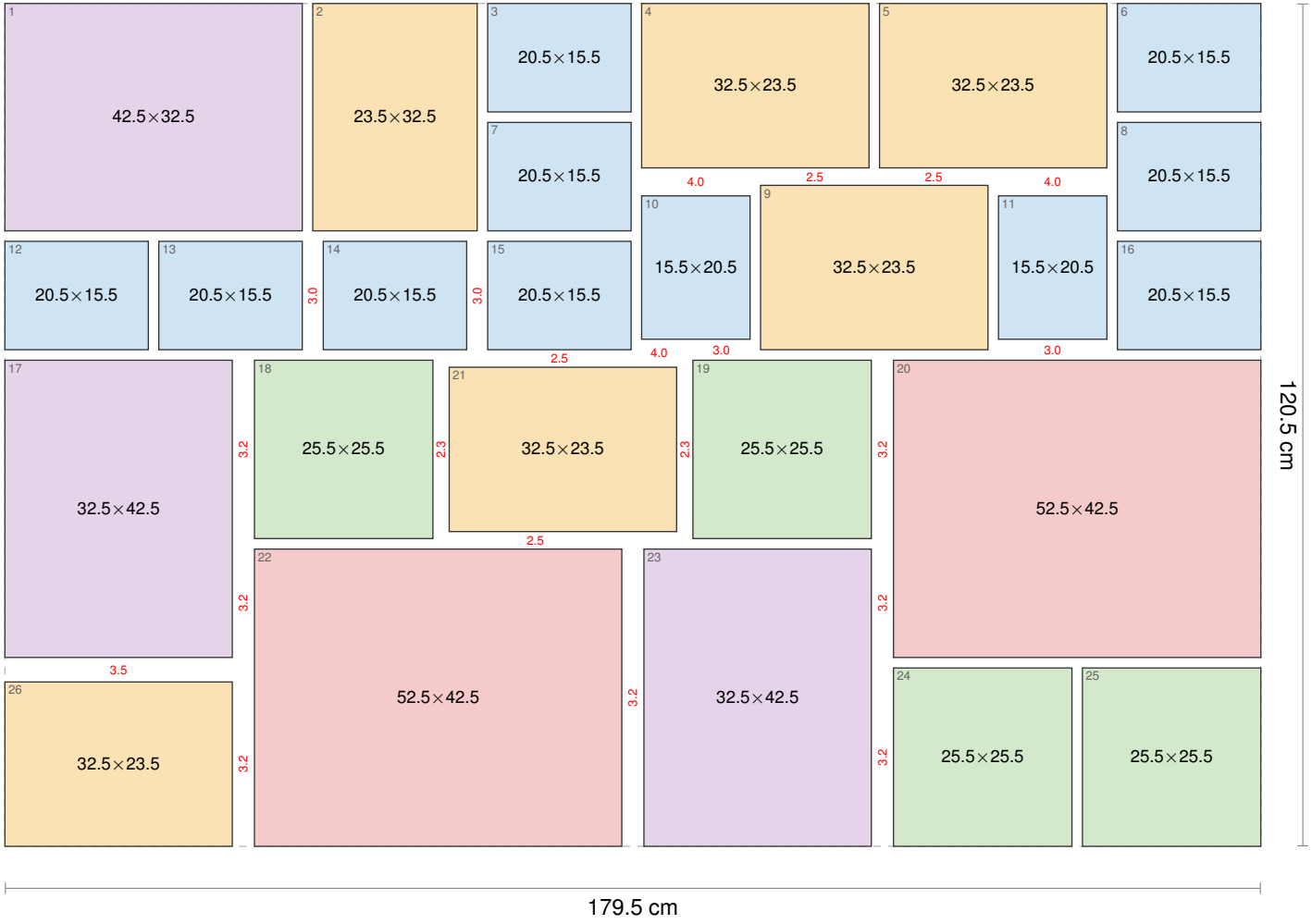
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.61 cm
Border notches	top 4 cm, bottom 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 104 cm
Entropy	0.77
mix / variety	0.83 / 0.73
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	36.0	0.0
3	23.5 × 32.5	90.0	0.0
4	20.5 × 15.5	115.0	0.0
5	20.5 × 15.5	137.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	115.0	17.0
8	42.5 × 32.5	137.0	17.0
9	20.5 × 15.5	91.5	34.0
10	20.5 × 15.5	115.0	34.0
11	25.5 × 25.5	0.0	44.0
12	25.5 × 25.5	63.0	44.0
13	32.5 × 23.5	28.0	45.0
14	32.5 × 23.5	90.0	51.0
15	52.5 × 42.5	127.0	51.0
16	20.5 × 15.5	0.0	71.0
17	32.5 × 23.5	39.0	71.0
18	15.5 × 20.5	22.0	72.5
19	15.5 × 20.5	73.0	72.5
20	32.5 × 42.5	90.0	78.0
21	20.5 × 15.5	0.0	88.0
22	25.5 × 25.5	127.0	95.0
23	25.5 × 25.5	154.0	95.0
24	32.5 × 23.5	22.0	97.0
25	32.5 × 23.5	56.0	97.0
26	20.5 × 15.5	0.0	105.0

Layout 16 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.60 cm
Border notches	bottom 10 cm, left 4 cm
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 95 cm
Entropy	0.77
mix / variety	0.83 / 0.78
stagger / lines	0.49 / 1.00 (1 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 32.5	0.0	0.0
2	23.5 × 32.5	44.0	0.0
3	20.5 × 15.5	69.0	0.0
4	32.5 × 23.5	91.0	0.0
5	32.5 × 23.5	125.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	69.0	17.0
8	20.5 × 15.5	159.0	17.0
9	32.5 × 23.5	108.0	26.0
10	15.5 × 20.5	91.0	27.5
11	15.5 × 20.5	142.0	27.5
12	20.5 × 15.5	0.0	34.0
13	20.5 × 15.5	22.0	34.0
14	20.5 × 15.5	45.5	34.0
15	20.5 × 15.5	69.0	34.0
16	20.5 × 15.5	159.0	34.0
17	32.5 × 42.5	0.0	51.0
18	25.5 × 25.5	35.7	51.0
19	25.5 × 25.5	98.3	51.0
20	52.5 × 42.5	127.0	51.0
21	32.5 × 23.5	63.5	52.0
22	52.5 × 42.5	35.7	78.0
23	32.5 × 42.5	91.3	78.0
24	25.5 × 25.5	127.0	95.0
25	25.5 × 25.5	154.0	95.0
26	32.5 × 23.5	0.0	97.0